Project proposal

AquaMonitor

Intelligent Multi-Parameter Water Quality Monitoring System

Empowering Domestic Water Safety & Aquaculture through Embedded Intelligence

SUBMITTED BY
Waithaka Njoroge

SPONSORING BODY
AFRETEC
African Engineering & Technology Network

DATE
April 2026
University of Nairobi

1. About the Proposer

My name is Waithaka Njoroge, a student at the University of Nairobi pursuing BSC IN MICROPROCESSOR TECHNOLOGY AND INSTRUMENTATION.

I am passionate about developing technology that solves urgent, tangible problems for communities across Africa. Water quality whether for drinking at the household level or for sustaining aquaculture enterprises is one of the most pressing challenges facing Kenya and the broader region. AquaMonitor is my response to that challenge: a precision, low-cost, IoT-enabled monitoring system designed from the ground up with African contexts in mind.

2. AFRETEC Sponsorship & Academic Context

This is my planned project after undertaking the a course on Embedded systems training which was sponsored by AFRETEC in collaboration with GEARBOX-EUROPLACER

AquaMonitor is being developed under the generous sponsorship of AFRETEC (the African Engineering and Technology Network) which supports African university students in undertaking impactful embedded systems research and engineering projects. AFRETEC's mission to cultivate the next generation of African technology innovators directly aligns with this project's goals.

This sponsorship provides not only financial support but also institutional legitimacy and mentorship enabling a structured, rigorous approach to the development lifecycle that goes beyond typical student projects. Under AFRETEC's framework, AquaMonitor will adhere to professional engineering standards in hardware design, firmware architecture, documentation, and commercialization planning.

Why This Project — My Motivation

Growing up in Kenya and being exposed to the realities of water insecurity in many places in kenya, I have witnessed firsthand how poor water quality silently endangers families who consume untested domestic water, and how aquaculture farmers suffer significant fish mortality losses because they lack affordable, realtime tools to monitor their pond conditions. These are not niche problems rather they affect millions of households and thousands of small-scale fish farmers across East Africa.

Existing water quality monitoring solutions are either prohibitively expensive laboratory-grade instruments costing hundreds of thousands of shillings, or fragmented single-parameter devices that give an incomplete picture. There is no affordable, integrated, multi-parameter water quality monitoring system designed specifically for the African market. AquaMonitor fills that gap.

By pursuing this project under AFRETEC sponsorship, I am gaining deep, hands-on experience in: embedded systems design and programming using both CircuitPython (rapid prototyping) and C (production firmware); sensor integration, signal conditioning, and analog-to-digital conversion; IoT connectivity and real-time data pipelines; and the full product commercialization journey from prototype to market. This experience is precisely the kind of transformative learning that AFRETEC champions.

3. Executive Summary

AquaMonitor is a smart, embedded water quality monitoring system built on the Raspberry Pi Pico W microcontroller platform. It simultaneously measures six critical water quality parameters — pH, salinity, dissolved oxygen (DO), temperature, turbidity, and overall purity index — and transmits the data wirelessly for real-time monitoring, logging, and alerting.

The system targets two primary markets: domestic households seeking assurance of safe drinking water, and aquaculture (fish and shrimp farming) operations where water parameters directly determine the productivity and survival rates of aquatic stock. AquaMonitor bridges the affordability gap by delivering laboratory-quality measurements at a fraction of the cost of existing solutions.

Prototype Platform Raspberry Pi Pico W (RP2040)	**Production Platform** Custom PCB — Seleste AquaMonitor v1.0(Currently unavailable)
Prototype Firmware CircuitPython	**Production Firmware** C (Pico SDK / bare-metal)
Parameters Monitored pH, Salinity, DO, Temperature, Turbidity, Purity Index	**Connectivity** Wi-Fi (802.11 b/g/n) via Pico W on-board CYW43439

4. Problem Statement

4.1 Domestic Water Safety

In Kenya, approximately 41% of the population lacks access to safely managed drinking water, according to WHO/UNICEF Joint Monitoring Programme data. Even households with piped or borehole water connections face contamination risks from pH imbalances, high turbidity, bacterial loading indicated by dissolved oxygen depletion, and excessive salinity none of which are detectable by sight, smell, or taste. Currently, water testing requires laboratory submissions that are time-consuming, expensive (KSh 2,000–15,000 per test), and inaccessible to most households. AquaMonitor puts continuous, real-time water safety verification directly in the home.

4.2 Aquaculture Productivity

Kenya's aquaculture sector has grown significantly, with over 1.2 million metric tonnes of fish farmed annually, yet productivity per pond remains far below global benchmarks. The primary reason is poor water quality management. Fish such as tilapia and catfish Kenya's dominant farmed species are acutely sensitive to dissolved oxygen levels (optimal: 5–8 mg/L), pH (optimal: 6.5–8.5), temperature (optimal: 25–30°C), and ammonia-induced turbidity increases. Without affordable continuous monitoring, farmers respond reactively to fish kills and disease outbreaks rather than proactively managing conditions. Losses from poor water quality management in Kenyan aquaculture are estimated at 15–30% of total stock value annually. AquaMonitor provides an affordable early warning system that can eliminate the majority of these losses.

5. Project Objectives

- Design and fabricate a multi-parameter water quality sensor node using the Raspberry Pi Pico W as the processing core.
- Implement prototype firmware in CircuitPython enabling rapid sensor validation, calibration routines, and Wi-Fi data transmission.
- Develop production-grade firmware in C using the Raspberry Pi Pico SDK, optimizing for low power consumption, real-time ADC sampling, and robust MQTT/HTTP communication.
- Integrate sensors for pH, salinity, dissolved oxygen, temperature, turbidity, and composite purity index measurement.
- Develop a cloud-connected dashboard for real-time parameter visualization, historical trending, threshold alerting, and SMS/email notifications.
- Validate the system in two field environments: a domestic household water storage tank and a live aquaculture pond.
- Produce a technical commercialization roadmap for scaling AquaMonitor to full production under the Seleste Technologies brand.

6. Significance, Merit & Impact

6.1 Impact on Domestic Water Safety

- Provides continuous 24/7 monitoring that replaces sporadic, expensive laboratory testing.
- Generates automatic safety alerts when parameters cross safe thresholds defined by Kenya Bureau of Standards (KEBS) and WHO guidelines.
- Enables communities, NGOs, and water utilities to maintain quality assurance records for regulatory compliance.
- Particularly impactful for rural boreholes and off-grid water systems where no monitoring infrastructure currently exists.

6.2 Impact on Aquaculture

- Early warning alerts for dissolved oxygen crashes (a leading cause of mass fish kills) allow farmers to intervene with aeration before losses occur.
- pH and temperature trending enables optimized feeding schedules and stocking density decisions.
- Salinity monitoring is critical for shrimp farming and coastal aquaculture operations.
- Turbidity tracking provides a proxy indicator for organic load and potential algal bloom development.
- Estimated reduction of fish loss by 20–30% through proactive parameter management, directly improving farmer income.

6.3 Academic & Engineering Merit

- Demonstrates end-to-end embedded systems engineering: sensor integration, analog signal conditioning, ADC, firmware, wireless communication, and cloud data pipelines.
- Explores the comparative advantages of interpreted scripting (CircuitPython) versus compiled embedded C for the same hardware target — a valuable pedagogical contribution.
- Provides a platform for future academic research in electrochemical sensor calibration drift, IoT security for embedded devices, and machine learning-based anomaly detection in water quality data streams.

7. Hardware Documentation — Prototype

7.1 Microcontroller — Raspberry Pi Pico W

The Raspberry Pi Pico W is the computational heart of the AquaMonitor prototype. It features the RP2040 dual-core ARM Cortex-M0+ processor running at up to 133 MHz, 264 KB SRAM, 2 MB onboard QSPI Flash, and critically an integrated Infineon CYW43439 chip providing IEEE 802.11 b/g/n Wi-Fi connectivity. Its 26 multi-function GPIO pins provide ample I/O for the full sensor suite, and three 12-bit ADC channels (plus one for the internal temperature sensor) handle analog sensor inputs directly.

Component	Qty	Purpose	Est. Cost (KSh)
Raspberry Pi Pico W	1	Main MCU — RP2040 dual-core + Wi-Fi	~KSh 1900
USB-A to Micro-USB Cable	1	Power & programming	~KSh 300
Half-size Breadboard (830 tie points)	2	Prototype circuit assembly	~KSh 400
Jumper Wire Kit (M-M, M-F, F-F)	1 set	Sensor wiring	~KSh 400
Micro-USB Power Supply (5V 2.5A)	1	Stable bench power	~KSh 800

7.2 Sensor Modules — Prototype

7.2.1 pH Sensor

A gravity-analog pH sensor module (compatible with DFRobot SEN0161 or equivalent) is used for prototype pH measurement. The module includes a BNC-connected glass pH electrode and a signal conditioning PCB that outputs a 0–3.3V analog voltage proportional to pH (0–14 scale). The Pico W reads this via its ADC. Calibration is performed using standard pH 4.0, 7.0, and 10.0 buffer solutions.

7.2.2 Salinity / Electrical Conductivity (EC) Sensor

Salinity is measured indirectly through electrical conductivity using a two-probe EC sensor module. The sensor passes a small AC signal between two platinum-coated probes immersed in water; conductivity (and thus salinity/TDS) is derived from the measured impedance. In prototype, a module such as the DFRobot Gravity EC Sensor (SEN0244) or a DIY conductivity cell with an op-amp signal conditioning circuit is used.

7.2.3 Dissolved Oxygen (DO) Sensor

Dissolved oxygen is measured using a galvanic or optical DO probe. The recommended prototype sensor is the DFRobot Gravity Dissolved Oxygen Sensor (SEN0237), which uses a galvanic probe producing a millivolt output proportional to DO concentration (0–20 mg/L). The analog output is temperature-compensated using readings from the DS18B20 temperature sensor. Calibration uses the air-saturation method.

7.2.4 Temperature Sensor

A waterproof DS18B20 digital temperature sensor probe is used, communicating via the Dallas 1-Wire protocol on a single Pico GPIO pin. This sensor provides 9–12 bit resolution over the range -55°C to +125°C with ±0.5°C accuracy more than sufficient for water quality monitoring and DO temperature compensation.

7.2.5 Turbidity Sensor

Turbidity is measured optically using an infrared LED emitter and phototransistor receiver pair. Light scattering and absorption by suspended particles reduces the received IR signal proportional to turbidity. A prototype module such as the DFRobot SEN0189 Turbidity Sensor outputs both analog voltage (0–4.5V) and a digital threshold signal. The analog output is read via the Pico ADC and mapped to NTU (Nephelometric Turbidity Units).

7.2.6 Water Purity Index (TDS Sensor)

Overall water purity is approximated using a Total Dissolved Solids (TDS) sensor module. TDS is closely related to electrical conductivity and provides a composite measure of dissolved minerals, salts, and organic compounds. A gravity TDS sensor (e.g., DFRobot SEN0244 or a custom resistive probe bridge) reads via the Pico ADC. The combined reading of TDS with pH and turbidity gives a composite 'purity index' score computed in firmware.

Sensor / Component	Qty	Specification	Est. Cost (KSh)
pH Sensor Module (Gravity Analog, BNC)	1	pH 0–14, 0–3.3V output	~KSh 3,500
EC/Salinity Sensor Module	1	0–20 mS/cm, analog output	~KSh 3,000
Dissolved Oxygen Probe + PCB (SEN0237)	1	0–20 mg/L, galvanic cell	~KSh 6,500
DS18B20 Waterproof Temp. Probe	2	1-Wire, ±0.5°C, −55–125°C	~KSh 600-1500
Turbidity Sensor Module (SEN0189)	1	0–3000 NTU, analog output	~KSh 2,500
TDS/Purity Sensor Module	1	0–1000 ppm, analog output	~KSh 1,200-2000
4.7kΩ Resistor (pull-up for 1-Wire)	2	DS18B20 bus termination	~KSh 10
100nF Decoupling Capacitors	6	ADC line filtering	~KSh 2
10kΩ Resistors (misc signal dividers)	6	Voltage divider / biasing	~KSh 10

7.3 Display, Power & Connectivity — Prototype

Component	Qty	Purpose	Est. Cost (KSh)
0.96" OLED Display (I2C, SSD1306)	1	128×64 px, local readout	~KSh 700-1500
Passive Buzzer	1	Audible threshold alerts	~KSh 50-150
3mm LEDs (Red, Green, Amber)	3	Status indicators	~KSh 30
3.7V LiPo Battery 2000 mAh	1	Portable field power	~KSh 1,200
TP4056 LiPo Charger Module	1	USB charging circuit	~KSh 200
IP67 Waterproof Project Enclosure	1	Sensor probe housing	~KSh 1,500
Cable Glands (PG7)	4	Watertight cable entry	~KSh 200

8. Hardware Documentation — Commercial Production

The commercial AquaMonitor product branded as AquaMonitor v1.0 transitions from the breadboard prototype to a custom-designed, professionally manufactured PCB. The Raspberry Pi Pico W remains the processing core in Version 1.0 (leveraging its integrated Wi-Fi and USB bootloader), mounted via castellated-edge SMD pads directly onto the AquaMonitor carrier board. Subsequent versions will explore direct RP2040 chip integration for further cost reduction.

8.1 Production PCB Specifications(*searched suggestions*)

Parameter	Value	Notes
PCB Layers	2-layer FR4	Cost-optimized; separate analog/digital ground planes
PCB Dimensions	80 × 60 mm	Compact form factor for enclosure integration
PCB Thickness	1.6 mm	Standard FR4 for mechanical rigidity
Surface Finish	HASL Lead-Free	Cost-effective, RoHS compliant
Minimum Track/Space	0.15 mm / 0.15 mm	JLCPCB / PCBWay compatible
MCU Footprint	Pico W castellated	Direct board-to-board soldering
Connector Type	BNC (pH, DO) + JST-XH (other sensors)	Industry-standard probe compatibility

Power Input	USB-C + 3.7V LiPo	Dual power path with auto-switchover

8.2 Production Component Selection

Component	Function	Key Specification
Raspberry Pi Pico W	MCU	RP2040 + CYW43439 Wi-Fi, castellated mount
Atlas Scientific EZO-pH Circuit	pH measurement	Industrial grade, UART/I2C interface, ±0.001 pH
Atlas Scientific EZO-EC Circuit	Conductivity/Salinity	UART/I2C, 0.07–500,000 µS/cm
Atlas Scientific EZO-DO Circuit	Dissolved Oxygen	UART/I2C, 0–100% saturation or 0.01–100+ mg/L
DS18B20 (TO-92 package, SMD adapt.)	Temperature	1-Wire, ±0.5°C, -55 to +125°C
AMSIL TSD-10 or custom IR pair	Turbidity	Transmissive optical, 0–1000 NTU
Custom Resistive TDS Bridge	TDS/Purity	Onboard conductivity cell, ADC measurement
SSD1306 OLED 128×64 (I2C)	Display	Onboard parameter readout
ESP-PROG UART header (3.3V TTL)	Debug/flash	Serial monitoring during production test
MCP3204 12-bit SPI ADC	Extra ADC channels	4-channel expansion for analog sensors
AMS1117-3.3 LDO Regulator	Power	5V → 3.3V, 800mA, SOT-223 package
TP4056 LiPo Charger IC	Battery mgmt	1A charge current, DW01A protection
CH340G USB-UART Bridge	USB serial	Firmware flashing & serial monitor via USB-C
Passive Components (R, C, L)	Various	Decoupling, filtering, biasing, ESD protection

9. Software & Firmware Implementation

9.1 Prototype Firmware :CircuitPython

CircuitPython is chosen for the prototype phase due to its rapid iteration cycle, extensive library ecosystem (Adafruit CircuitPython libraries), and ease of debugging via USB serial REPL. The Raspberry Pi Pico W is fully supported by

CircuitPython 9.x, which includes built-in Wi-Fi support through the 'wifi' and 'socketpool' modules.

9.1.1 Prototype Firmware Architecture

- main.py — Main execution loop: sensor reads, computation, display update, Wi-Fi transmission, alert logic
- sensors/ph_sensor.py — pH ADC read, moving average filter, 3-point calibration curve (pH 4/7/10)
- sensors/ec_sensor.py — Conductivity measurement, temperature compensation, salinity conversion
- sensors/do_sensor.py — DO ADC read, Benson & Krause temperature-compensation algorithm
- sensors/turbidity_sensor.py — Turbidity ADC read, NTU conversion curve
- sensors/tds_sensor.py — TDS ADC read, ppm computation, purity index scoring
- sensors/temperature.py — DS18B20 1-Wire read via adafruit_onewire and adafruit_ds18x20 libraries
- display/oled.py — SSD1306 display management, parameter cycling screen layout
- comms/wifi_manager.py — Wi-Fi connection management with auto-reconnect
- comms/mqtt_client.py — MQTT publish to broker (HiveMQ Cloud or local Mosquitto)
- config.py — Centralized constants: Wi-Fi credentials, MQTT topic structure, alert thresholds

9.1.2 Key CircuitPython Libraries

- adafruit_onewire, adafruit_ds18x20 — DS18B20 temperature sensor
- adafruit_ssd1306 — OLED display driver
- adafruit_minimqtt — Lightweight MQTT client for Wi-Fi
- busio, analogio, digitalio — Core CircuitPython hardware interfaces

9.1.3 Calibration Workflow (CircuitPython)

Calibration is implemented as an interactive serial REPL menu accessible via USB. For pH: the probe is dipped in standard buffer solutions and offset/slope coefficients are computed and stored in non-volatile storage (microcontroller NVM or a JSON config file on the CIRCUITPY drive). DO calibration uses the air-saturation (100% DO) and sodium sulfite zero-DO method. EC calibration uses a known-conductivity standard solution. Calibration coefficients persist across power cycles.

9.2 Production Firmware — C (Raspberry Pi Pico SDK)

Production firmware is written in C using the official Raspberry Pi Pico SDK (based on CMake build system and ARM GCC toolchain). The transition to C delivers significantly improved performance, deterministic real-time behavior, and substantially lower power consumption through aggressive use of the RP2040's sleep modes. Production firmware supports over-the-air (OTA) firmware updates via the Wi-Fi connection.

9.2.1 Production Firmware Architecture

- main.c — System initialization, FreeRTOS task scheduler start
- tasks/sensor_task.c — 500ms periodic ADC sampling across all sensors; writes to shared ring buffer
- tasks/comms_task.c — MQTT/HTTPS publish; handles Wi-Fi reconnection; implements TLS 1.2 for secure transport
- tasks/display_task.c — OLED update at 1Hz; parameter scrolling; alert state display
- tasks/alert_task.c — Threshold comparison; GPIO buzzer/LED control; SMS trigger via HTTP API
- drivers/ph_driver.c — ADC read with oversampling (16x), IIR low-pass filter, Nernst equation calibration
- drivers/ec_driver.c — AC excitation waveform generation (PWM), conductivity bridge measurement
- drivers/do_driver.c — Galvanic cell ADC read, Benson temperature compensation table lookup
- drivers/ds18b20.c — Bit-banged 1-Wire master; ROM search; 12-bit conversion trigger & read
- drivers/turbidity.c — ADC oversampling, NTU lookup table from calibration polynomial
- drivers/tds.c — Resistive bridge ADC, ppm computation, purity scoring algorithm
- storage/nvs.c — Non-volatile storage in RP2040 flash (last 4KB sector); calibration & config persistence
- ota/ota_manager.c — HTTPS firmware image download, CRC32 validation, flash write & reboot
- CMakeLists.txt — Build configuration, linking FreeRTOS, lwIP (TCP/IP), mbedTLS (TLS)

9.2.2 RTOS & Middleware Stack(*future advancement*)

Library / Framework	Role	Notes
FreeRTOS	Real-time OS	task scheduling, queues, etc through guided indtructions
lwIP	TCP/IP Stack	Lightweight IP stack for Wi-Fi networking
mbedTLS	TLS/SSL	Encrypted HTTPS/MQTTS communication

MQTT (Paho C client)	Messaging Protocol	Publish sensor data to MQTT broker
Pico SDK ADC, PWM, PIO	Hardware Drivers	Native SDK peripherals — no HAL overhead

9.3 Cloud & Dashboard Architecture

AquaMonitor will be hosted to cloud server (after funding to acquire online hosting) and will serve simple web dashboard showing all the readings for all the sensors

10. Implementation Roadmap

Phase	Milestone	Deliverables & Activities
Phase 1 (Weeks 1–2)	Hardware Assembly & Sensor Bring-up	Assemble Pico W + all sensors on breadboard. CircuitPython bring-up. Verify raw ADC readings from all sensors. Identify signal conditioning needs.
Phase 2 (Weeks 3–4)	Sensor Calibration & Characterization	Implement and validate all sensor calibration routines. Compare readings against certified reference instruments. Document accuracy and precision.
Phase 3 (Weeks 5–6)	Prototype Firmware Complete	Complete CircuitPython firmware: sensor fusion, purity index algorithm, OLED display, Wi-Fi MQTT transmission, threshold alerting. First functional prototype.
Phase 4 (Weeks 7–8)	Cloud Dashboard Development	Build simple dashboard and Field testing at domestic water source and aquaculture pond.
Phase 5 (Weeks 9–11)	Production Firmware — C	Port all firmware to C (Pico SDK + FreeRTOS). Validate equivalent functionality. Optimize power consumption. Implement OTA update system.
Phase 6 (Weeks 12–14)	PCB Design & Fabrication	Design AquaMonitor v1.0 PCB throug guided instructuions
Phase 7 (Weeks 15–16)	Field Validation & Documentation	Extended field deployment. Accuracy benchmarking. Final technical report, user manual, and commercialization plan preparation.

11. Commercialization Strategy

AquaMonitor is designed from inception with commercialization as a first-class objective, not an afterthought.

11.1 Target Market Segments

- Domestic households with boreholes, water tanks, or piped connections who want water safety assurance, estimated 3.5 million households in Kenya alone.
- Small-to-medium aquaculture farmers (0.5–5 acre ponds) who currently have no affordable monitoring solution, estimated 50,000+ registered fish farmers in Kenya.
- Water utilities, county governments, and NGOs managing community water points and needing compliance monitoring data.
- Schools and hospitals requiring routine water quality documentation for regulatory purposes.

11.2 Competitive Advantage

- Africa-first design: calibration protocols, dashboard UI, and alert systems designed for Kenyan and East African regulatory standards and connectivity realities.
- Integrated multi-parameter platform at a price point 5–10x lower than comparable international instruments.
- Local support and service network through AFRETEC and GEARBOX programs

12. Proposed Budget

Budget Item	Notes
Prototype Sensors & Components	Full sensor suite, breadboard, passive components
Raspberry Pi Pico W (x3 units)	Prototype + spares + production validation unit
PCB Fabrication (10 units,)	AquaMonitor v1.0 PCBs with SMT assembly
Atlas Scientific EZO Modules (x3)	Production-grade pH, EC, and DO circuits
Enclosures, Cables & Connectors	Waterproof housing, BNC cables, cable glands
Cloud Infrastructure (6 months)	(Safaricom Hosting)
Laboratory Calibration Solutions	pH buffer sets, EC standards, DO calibration kit
Miscellaneous & Contingency (10%)	Component failures, shipping, unexpected needs

13. Conclusion

AquaMonitor represents a convergence of urgent social need, technical ambition, and commercial opportunity. By leveraging the power of the Raspberry Pi Pico W, a carefully selected multi-parameter sensor suite, and modern cloud and IoT infrastructure, this project will deliver a working, validated water quality monitoring system that can genuinely improve lives both in homes and on fish farms across Kenya and East Africa.

Under the sponsorship and mentorship of AFRETEC in collaboration with GEARBOX this project will also serve as a rigorous embedded systems education experience one that spans the full engineering lifecycle from breadboard prototype to commercial PCB, from interpreted scripting to production C firmware, and from sensor calibration to cloud dashboards. The skills, knowledge, and artifacts produced will be of lasting value to my academic and professional development.

I am confident that AquaMonitor, will become a commercially viable product that demonstrates what African university students with the right support and resources can achieve in the global embedded systems and IoT space. I am grateful for AFRETEC's vision in sponsoring this work and look forward to delivering a project that does justice to that investment.

Respectfully submitted by:

Waithaka Njoroge

Student, B.TECH
University of Nairobi
April 2026

Sponsored by:

AFRETEC
African Engineering & Technology Network
Embedded Systems Research Program
University of Nairobi